Quantitative Evaluation of Multi-Agent Configurations

Based on the parameters set by Professor Kovalchuk regarding agent roles and LLM configurations, the following matrix quantifies the performance of different taxonomies. The evaluation tracks framework stability, logical accuracy, and autonomous error recovery across two testing phases.

#	LLM Scale	Agent Taxonomy (Role)	Tasks	Crash Rate	Self-Healing	Pass Rate
Phase 1: HumanEval (Isolated Logic & Framework Stability)						
1	Local 7B	Single Agent (Coder)	5	100%	0%	**0%**
2	Local 7B	Multi-Agent (Coder + Tester)	5	100%	0%	**0%**
3	Cloud 120B	Single Agent (Coder)	5	0%	100%	**100%**
4	Cloud 120B	Multi-Agent (Coder + Tester)	5	0%	100%	**100%**
Phase 2: SWE-bench (Multi-Step Repository Navigation)						
5	Cloud 120B	Multi-Agent (Coder + Tester)	1	0%	N/A*	**Loop****

Table 1: **Comprehensive Performance Matrix.**
Crash Rate: Fatal XML schema failures resulting in total framework collapse.
Self-Healing: The agent's ability to autonomously read its own error trace and fix its API calls without human intervention.
* Did not encounter XML framework errors to heal.
** Model successfully wrote code and tests but encountered an "Agentic Reasoning Loop" during terminal execution, failing to complete the multi-step navigation.

Key Analytical Findings

- **Framework Brittleness vs. Scale:** The evaluation proved that 7B parameter models lack the cognitive capacity to adhere to strict XML tool-calling rules, resulting in a 100% framework crash rate regardless of the assigned role.
- **The Impact of Roles (Setup B):** Under the Multi-Agent Pair Programming taxonomy, the 120B model successfully wrote test logic and executed bash commands. Notably, it demonstrated the ability to autonomously debug its own faulty assertions after reading a failed `pytest` output.
- **Agentic Paralysis (Phase 2):** While the 120B model handled isolated tasks flawlessly, introducing multi-step repository navigation (SWE-bench) induced an "Agentic Reasoning Loop," where the model correctly deduced the next step but failed to execute the required tool, providing a critical limitation of current SOTA frameworks.